ModelIt!

Student Activity Pack

Engineering a Solution: The Design Process

Modeling How Iteration Turns Failure into Innovation

Grade 06L2

NGSS MS-ETS1-1, MS-ETS1-2

Influence of Engineering, Technology, and Science on Society

STUDENT NAME _____

DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Have you ever built something that did not work the first time? What did you do next?

2 Why do you think engineers test their designs before producing the final product?

3 Is failure a bad thing in engineering? Why or why not?

4 Draw the steps you think engineers follow when designing a new product.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Problem Complexity | Research Depth | Design Quality | Prototype Performance | Iteration Count

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: No Research Rush

High complexity, no research

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Research First

High complexity, deep research

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Iterate to Excellence

Moderate research, observe iterations

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Budget Constraint — Money available for materials and testing. Lower budget limits options and forces creative trade-offs

Team Collaboration — How effectively team members share ideas, divide tasks, and integrate feedback. Better collaboration improves outcomes

Time Pressure — Deadline constraints that limit how many iterations are possible. More time allows more improvement cycles

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do the best engineering solutions almost never work on the first try, and how does failure actually make designs better? Use evidence from your simulation to support your answers.

1. **Why is a tested-and-improved second design usually better than even a carefully planned first design?**

2. **How does the feedback loop between testing and redesign create a cycle of improvement?**

3. **What would happen if an engineer built their first design and shipped it without testing?**

4. **How is the engineering design process similar to how scientists do experiments?**

5. **Why does more research before designing lead to a better FIRST prototype?**

8 STEM Challenge: The Iterative Design Challenge

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YOUR ENGINEERING MISSION

Design, test, and iterate on a solution to a real engineering problem, documenting how each iteration improves performance.

THE SCENARIO

Your engineering firm has been hired to design a device that protects a raw egg from a 3-meter drop using only limited materials. You have three test rounds. After each test, you must analyze what failed, redesign, and test again. The team with the most improvement across iterations wins.

Engineering Design Thinking

1. WHAT ARE THE CRITERIA (MUST PROTECT THE EGG) AND CONSTRAINTS (LIMITED MATERIALS, 3-METER DROP HEIGHT)?
2. WHAT SCIENTIFIC PRINCIPLES (FORCE DISTRIBUTION, ENERGY ABSORPTION, DECELERATION) WILL GUIDE YOUR DESIGN?
3. HOW WILL YOU DOCUMENT WHAT YOU LEARN FROM EACH TEST TO INFORM THE NEXT ITERATION?
4. WHAT TRADE-OFFS ARE YOU MAKING IN YOUR DESIGN (WEIGHT VS. PROTECTION, SIMPLICITY VS. EFFECTIVENESS)?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Engineering Design Specialists and Product Developers use iterative prototyping and testing to create everything from medical devices to consumer electronics to spacecraft. They earn

$65,000-$130,000/year.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In the engineering design model, what happens when a team jumps into building a prototype for a complex problem without doing any research first?

- **A.** The prototype works well because hands-on building is more effective than research
- **B.** Design quality is low because the team does not understand the problem well enough, leading to poor prototype performance
- **C.** Research is unnecessary because testing will reveal everything they need to know
- **D.** The prototype performs the same regardless of research depth

2 Why is iteration the single most powerful variable for improving engineering outcomes?

- **A.** Iteration allows engineers to start over from scratch with each attempt
- **B.** Each test-improve cycle reveals problems that planning alone cannot predict, and fixing those problems creates a positive feedback loop of improvement
- **C.** Iteration is just repeating the same design until it works by luck
- **D.** More iterations mean more materials are used, which always produces better results

3 The model shows a positive feedback loop between prototype testing and design quality. What does this mean?

- **A.** Positive feedback means the design gets worse with each test
- **B.** Testing a prototype provides specific information about what works and what fails, which directly improves the next design, which produces an even better prototype
- **C.** Positive feedback means the engineer feels more confident after each test
- **D.** The prototype automatically improves itself without human intervention

4 Two teams tackle the same engineering problem. Team A researches thoroughly and builds one prototype. Team B does minimal research but builds and tests three prototypes. Based on the model, which team likely produces a better solution?

- **A.** Team A, because research always produces better results than testing
- **B.** Team B, because three iterations of testing and improvement typically outperform one well-researched attempt that has not been tested against reality
- **C.** Both teams produce identical results because the problem is the same
- **D.** Neither team can succeed without combining research and iteration

5 **How does increasing problem complexity affect the engineering design process?**

A. More complex problems require fewer iterations because engineers work harder

B. More complex problems produce worse first designs because more constraints and criteria interact in unpredictable ways, making research and iteration even more important

C. Problem complexity has no effect because all engineering follows the same steps

D. More complex problems are easier because they have more possible solutions